

GENERAL SCIENCE, Paper-I
(Physical Science)
(English Version)

Time : 2 Hours 45 Mins.]

[Maximum Marks : 50

Instructions :

1. There are four sections and 33 questions in this paper.
 2. Answer should be written in a given answer booklet.
 3. There is internal choice in Section – IV.
 4. Write all the Questions visible and legibly.
 5. 15 minutes are given for reading the question paper and 2 hours 30 mins. for answering questions.
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Section – I**Note :** (1) Answer all the questions.**12 × ½ = 6**

(2) Each question carries ½ mark.

1. Three bodies A, B and C are in thermal equilibrium. The temperature of B is 50 °C. Then the temperature of C is
(A) 55 °C (B) 50 °C (C) 45 °C (D) 40 °C
2. A solution turns red litmus into blue, its pH value is _____.
(A) 1 (B) 4 (C) 5 (D) 10
3. **Statement P :** Optically denser medium may not possess greater mass density.
Statement Q : Kerosene with high refractive Index is optically denser than water.
(A) Both P and Q are correct. (B) P-correct, Q-wrong.
(C) P-wrong, Q-correct. (D) Both P and Q are wrong.
4. Which of the following material cannot be used to make a lens ?
(A) Water (B) Glass (C) Plastic (D) Clay
5. The value of least distance of distinct vision of the healthy human beings _____.
6. Which rule is violated in the electronic configuration $1s^0 2s^2 2p^4$?

7. Noble gases belong to _____ group of modern periodic table.

8. Match the following :

Molecule		Bond angle	
(i) Ammonia	()	(p)	$104^{\circ}31'$
(ii) Boron trifluoride	()	(q)	$107^{\circ}48'$
(iii) Water	()	(r)	120°
(A) (i)-(p), (ii)-(r), (iii)-(q)			
(B) (i)-(q), (ii)-(p), (iii)-(r)			
(C) (i)-(q), (ii)-(r), (iii)-(p)			
(D) (i)-(p), (ii)-(q), (iii)-(r)			

9. Which element is not having 8 electrons in its valency shell among noble gases ?

Answer 10 and 11 questions based on below table :

Material	Silver	Iron	Drinking Water	Air
Specific Resistance ($\Omega\text{-m}$) at 20°C	1.59×10^{-8}	1×10^{-7}	2×10^{-1}	1.3×10^{16}

10. In which material the electric current is more ?

11. What is the SI unit of Specific Resistance ?

12. The impurity present in the ore is called as _____.

- (A) Gangue (B) Flux (C) Slag (D) Mineral

Section – II

Note : (1) Answer all the questions.

$8 \times 1 = 8$

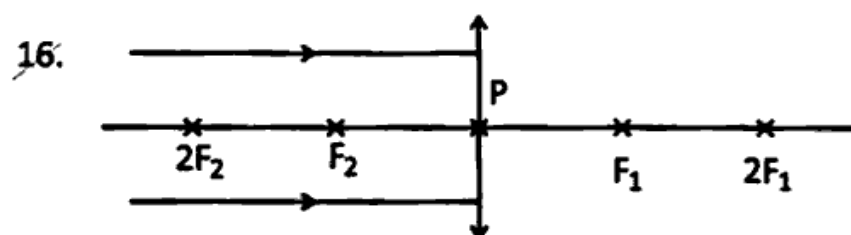
(2) Each question carries 1 mark.

13. Convert 27°C into Kelvin scale.

14. Refractive index of glass relative to water is $\frac{9}{8}$. What is the refractive index of water relative to glass ?

15. Write any two material required in the activity.

"To find the refractive index of a glass slab."



Complete the ray diagram with appropriate refracted rays.

17. An electron in an atom has the following set of four quantum numbers of $3s^1$:

n	l	m_l	m_s
3	0	0	$+\frac{1}{2}$

Then write four quantum numbers for $2s^1$ electron.

18. Write the Mendeleef's Periodic law.
19. An element X belongs to 3rd period and group 2 of the modern periodic table. Predict the number of valence electrons and write.
20. Mention the daily life application of thermite process.

Section – III

Note : (1) Answer all the questions.

8 × 2 = 16

(2) Each question carries 2 marks.

21. Your friend is asked to differentiate between evaporation and boiling. What questions would you ask to make him to know the differences between evaporation and boiling ?
22. Write any two daily life uses of lenses.
23. How do you appreciate the work of ciliary muscles in the eye ?
24. Write the material required to prove Ohm's law activity.
25. Fill the table given below :

Baking Soda	Washing Soda	
NaHCO_3		$\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$

26. Explain the terms in nI^x method.
27. Predict the reasons for low melting point for covalent compounds when compared with ionic compounds.
28. Write the names of any two ores of Iron.

Section – IV

Note : (1) Answer all the questions.

5 × 4 = 20

(2) Each question carries 4 marks.

(3) Every Question has Internal Choice.

29. What is Myopia ? Explain the correction of the defect Myopia.

OR

Deduce the expression for the equivalent resistance of three resistors R_1 , R_2 , R_3 ohms connected in series.

30. How do the following properties change in a group and period ?

- (i) Atomic radius
- (ii) Ionization energy
- (iii) Electron affinity
- (iv) Electronegativity

OR

Explain the formation of the O_2 molecule using valence bond theory.

31. Suggest an experiment to find the specific heat of solid.

OR

How do you verify experimentally that $\frac{\sin i}{\sin r}$ is a constant ?

32. Observe the table and answer the following questions :

Solution	A	B	C	D	E	F	G	H
pH Value	8	2	6	7	13	1	9	12

(i) Which solution is neutral ?

(ii) Which solutions are strong acids ?

(iii) Which solutions are strong bases ?

(iv) Which solutions are weak bases ?

OR

Complete the following table :

Element	Atomic Number	Electronic Configuration
Carbon	6	$1s^2 2s^2 2p^2$
Oxygen	8	
Sodium	11	
Argon	18	
Calcium	20	

33. Draw ray diagrams for the following positions with respect to convex lens :

(i) Object is placed beyond $2F_2$.

(ii) Object is placed between $2F_2$ and F_2 .

OR

Draw a neat diagram of Reverberatory furnace.